Kyle Genova

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EXPERIENCE

Google DeepMind — Staff Research Scientist Oct. 2025 – Present
Frontier AI Unit *New York, NY*

- Core contributor to Gemini, Gemini Omni, and Genie training.
- Expertise in 3D understanding and generation. Experience designing experiments, training very large models, and contributing very large quantities of data. Pre-training run captain.
- First author for multiple approved Gemini RFCs (20+ contributors). Designed and ran experiments leading to improvements in Gemini's spatial understanding capabilities.
- Own the majority of the 3D training data mixture for an upcoming model training effort.

Google DeepMind — Senior Research Scientist May 2024 – Oct. 2025
Foundational Research Unit *New York, NY*

- Led Geo's Foundational Features project, a 2-year 3D model training effort to build a world-scale 3D feature representation enabling open-vocabulary scene-understanding queries for the Google Maps team.
- Trained research models in support of Gemini 3, authored approved changes shown to improve its depth, correspondence, detection, and geospatial capabilities.
- Published academic papers across top conferences, mentored 5+ student researcher projects.

Google Research — Research Scientist May 2021 – May 2024
Machine Perception — promoted to Senior Research Scientist, Oct. 2023 *Mountain View, CA & New York, NY*

- Published and productionized 2D3DNet, the most used 3D understanding model across Google Maps since 2021 — driving over 30 million map updates and supporting routing, HD maps, 3D reconstruction, and indoor Immersive View. Over one quadrillion 3D points served.

Google Research — Research Intern (5 internships) 2016 – 2021
New York, Cambridge, and Mountain View

- Research projects in 3D vision, graphics, and machine learning; 9 publications and 4 patents.

EDUCATION

Princeton University — Ph.D., Computer Science 2016 – 2021
Advisor: Prof. Thomas Funkhouser *Princeton, NJ*

- Thesis: *3D Representations for Learning to Reconstruct and Segment Shapes*.

Cornell University — B.A., Computer Science 2012 – 2016
GPA 4.17/4.30; Phi Beta Kappa (top 3% of class) *Ithaca, NY*

SELECTED PUBLICATIONS

24 papers • **6,000+** citations • h-index **22** • CVPR **Best Paper Nominee** • full list on Google Scholar

- Vision Banana: Image Generators are Generalist Vision Learners. *Valentin Gabeur, Shangbang Long, Songyou Peng, Paul Voigtlaender, …, **Kyle Genova**, …, Kaiming He, Thomas Funkhouser, Jean-Baptiste Alayrac, Radu Soricut (25 authors)*. arXiv, 2026
- CityRAG: Stepping Into a City via Spatially-Grounded Video Generation. *Gene Chou, Charles Herrmann, **Kyle Genova**, Boyang Deng, Songyou Peng, Bharath Hariharan, Jason Y. Zhang, Noah Snavely, Philipp Henzler*. arXiv, 2026
- MoMaps: Semantics-Aware Scene Motion Generation with Motion Maps. *Jiahui Lei, **Kyle Genova**, George Kopanas, Noah Snavely, Leonidas Guibas*. ICCV 2025
- Visual Chronicles: Using Multimodal LLMs to Analyze Massive Collections of Images. *Boyang Deng, Songyou Peng, **Kyle Genova**, Gordon Wetzstein, Noah Snavely, Leonidas Guibas, Thomas Funkhouser*. ICCV 2025

- SplatTalk: 3D VQA with Gaussian Splatting. *Anh Thai, Songyou Peng, **Kyle Genova**, Leonidas Guibas, Thomas Funkhouser*. ICCV 2025
- NIFTY: Neural Object Interaction Fields for Guided Human Motion Synthesis. *Nilesh Kulkarni, Davis Rempe, **Kyle Genova**, Abhijit Kundu, Justin Johnson, David Fouhey, Leonidas Guibas*. CVPR 2024
- Nerflets: Local Radiance Fields for Efficient Structure-Aware 3D Scene Representation from 2D Supervision. *Xiaoshuai Zhang, Abhijit Kundu, Thomas Funkhouser, Leonidas Guibas, Hao Su, **Kyle Genova***. CVPR 2023
- OpenScene: 3D Scene Understanding with Open Vocabularies. *Songyou Peng, **Kyle Genova**, Chiyu "Max" Jiang, Andrea Tagliasacchi, Marc Pollefeys, Thomas Funkhouser*. CVPR 2023
- Panoptic Neural Fields: A Semantic Object-Aware Neural Scene Representation. *Abhijit Kundu, **Kyle Genova**, Xiaoqi Yin, Alireza Fathi, Caroline Pantofaru, Leonidas Guibas, Andrea Tagliasacchi, Frank Dellaert, Thomas Funkhouser*. CVPR 2022
- Learning 3D Semantic Segmentation with only 2D Image Supervision. ***Kyle Genova**, Xiaoqi Yin, Abhijit Kundu, Caroline Pantofaru, Forrester Cole, Avneesh Sud, Brian Brewington, Brian Shucker, Thomas Funkhouser*. 3DV 2021 (**Oral**)
- IBRNet: Learning Multi-View Image-Based Rendering. *Qianqian Wang, Zhicheng Wang, **Kyle Genova**, Pratul P. Srinivasan, Howard Zhou, Jonathan T. Barron, Ricardo Martin-Brualla, Noah Snavely, Thomas Funkhouser*. CVPR 2021
- Local Deep Implicit Functions for 3D Shape. ***Kyle Genova**, Forrester Cole, Avneesh Sud, Aaron Sarna, Thomas Funkhouser*. CVPR 2020 (**Oral**)
- CvxNet: Learnable Convex Decomposition. *Boyang Deng, **Kyle Genova**, Soroosh Yazdani, Sofien Bouaziz, Geoffrey Hinton, Andrea Tagliasacchi*. CVPR 2020 (**Best Paper Nominee**)
- Towards Fairness in Visual Recognition: Effective Strategies for Bias Mitigation. *Zeyu Wang, Klint Qinami, Ioannis Christos Karakozis,**Kyle Genova**, Prem Nair, Kenji Hata, Olga Russakovsky*. CVPR 2020
- Learning Shape Templates with Structured Implicit Functions. ***Kyle Genova**, Forrester Cole, Daniel Vlasic, Aaron Sarna, William T. Freeman, Thomas Funkhouser*. ICCV 2019
- Text-based Editing of Talking-head Video. *Ohad Fried, Ayush Tewari, Michael Zollhöfer, Adam Finkelstein, Eli Shechtman, Dan B. Goldman, **Kyle Genova**, Zeyu Jin, Christian Theobalt, Maneesh Agrawala*. SIGGRAPH 2019
- Unsupervised Training for 3D Morphable Model Regression. ***Kyle Genova**, Forrester Cole, Aaron Maschinot, Aaron Sarna, Daniel Vlasic, William T. Freeman*. CVPR 2018 (**Spotlight**)
- An Experimental Evaluation of the Best-of-Many Christofides' Algorithm for the Traveling Salesman Problem. ***Kyle Genova**, David P. Williamson*. Algorithmica 2017 (**ESA Selected Publication**); ESA 2015

PATENTS (6)

- Neural semantic fields for generalizable semantic segmentation of 3D scenes. *US 12,602,898 (2026)*.
- Systems and methods for generating splat-based differentiable two-dimensional renderings. *US 12,288,295 (2025)*.
- Convex representation of objects using neural network. *US 11,978,268 (2024); US 11,508,167 (2022)*.
- Learning to reconstruct 3D shapes by rendering many 3D views. *US 10,510,180 (2019); US 10,403,031 (2019)*.

ACADEMIC SERVICE

- **Area Chair:** CVPR 2026 (*Outstanding Area Chair*); ICCV 2025; SIGGRAPH 2026 (Technical Papers Committee).
- **Committees:** SIGGRAPH Asia 2026 Technical Workshops Committee.
- **Reviewer:** CVPR, ICCV, ECCV, NeurIPS, SIGGRAPH, and SIGGRAPH Asia since 2018 — four *Outstanding/Top Reviewer* recognitions (CVPR 2021, ECCV 2020, NeurIPS 2019 & 2025).

HONORS & AWARDS

- Siebel Scholar, Class of 2021 — The Thomas and Stacey Siebel Foundation, 2020.
- NSF Graduate Research Fellowship — National Science Foundation, 2018–2021.
- Graduate Student Teaching Award — Princeton University, 2018.
- Gordon Y.S. Wu Fellowship in Engineering — Princeton University, 2016–2021.
- Hertz Foundation Top 120 Candidates, 2016.
- Phi Beta Kappa; CRA Outstanding Undergraduate Researcher Nomination — Cornell University, 2015.

TEACHING & MENTORING

- **Mentoring:** Hosted or co-hosted five research interns; mentored work published at top vision venues (ICCV, CVPR).
- **Teaching:** Princeton University, Assistant in Instruction — Computer Graphics (COS 426), Computer Vision (COS 429). *Graduate Student Teaching Award.*